Inpatient claim review assistant

The system in one sentence

Help staff review hospital claims with checked rules, clear evidence and an owned next step.

Key design calls

Publish drafts for staff. Rejected: Automatic settlement.. Why: Payment authority is unknown.

Code applies dated rules. Rejected: Model-made limits.. Why: Coverage must be repeatable.

Check each source quote. Rejected: Trust fluent explanations.. Why: Staff need proof.

Read back uncertain saves. Rejected: Blind retry.. Why: Avoid duplicate records.

Who acts and who decides

Action	Autonomy
Read approved claim files	acts alone
Explain evidence	suggests
Set eligible charges	acts alone
Resolve exceptions and rejections	human decides
Settle claims	human decides

Top risks by RPN

FM2 | Wrong coverage or room adjustment | RPN 225

FM8 | Human queue silently stalls | RPN 200

FM3 | Forged citation or malicious document | RPN 180

FM5 | Uncertain save lost or repeated | RPN 160

FM4 | Duplicate/concurrent drafts confuse adjudication | RPN 128

Launch blockers and owners

H1: First release publishes staff-facing recommendations... Owner: Claims director.

H2: Insurer supplies signed, effective-dated policy... Owner: Insurance rules owner.

H3: Privacy owner selects deployment region... Owner: Privacy officer.

H4: Claims team covers a 24-hour... Owner: Claims operations lead.

Illustrative cost per case

USD 0.02112 per case in the model budget. This is an estimate, not a current provider quote. Staff and tools cost extra.

Review status: independent-agent; round 1; needs-work.

Proposed design. Full rules and blockers: design.json and assumptions.md.

Task tree

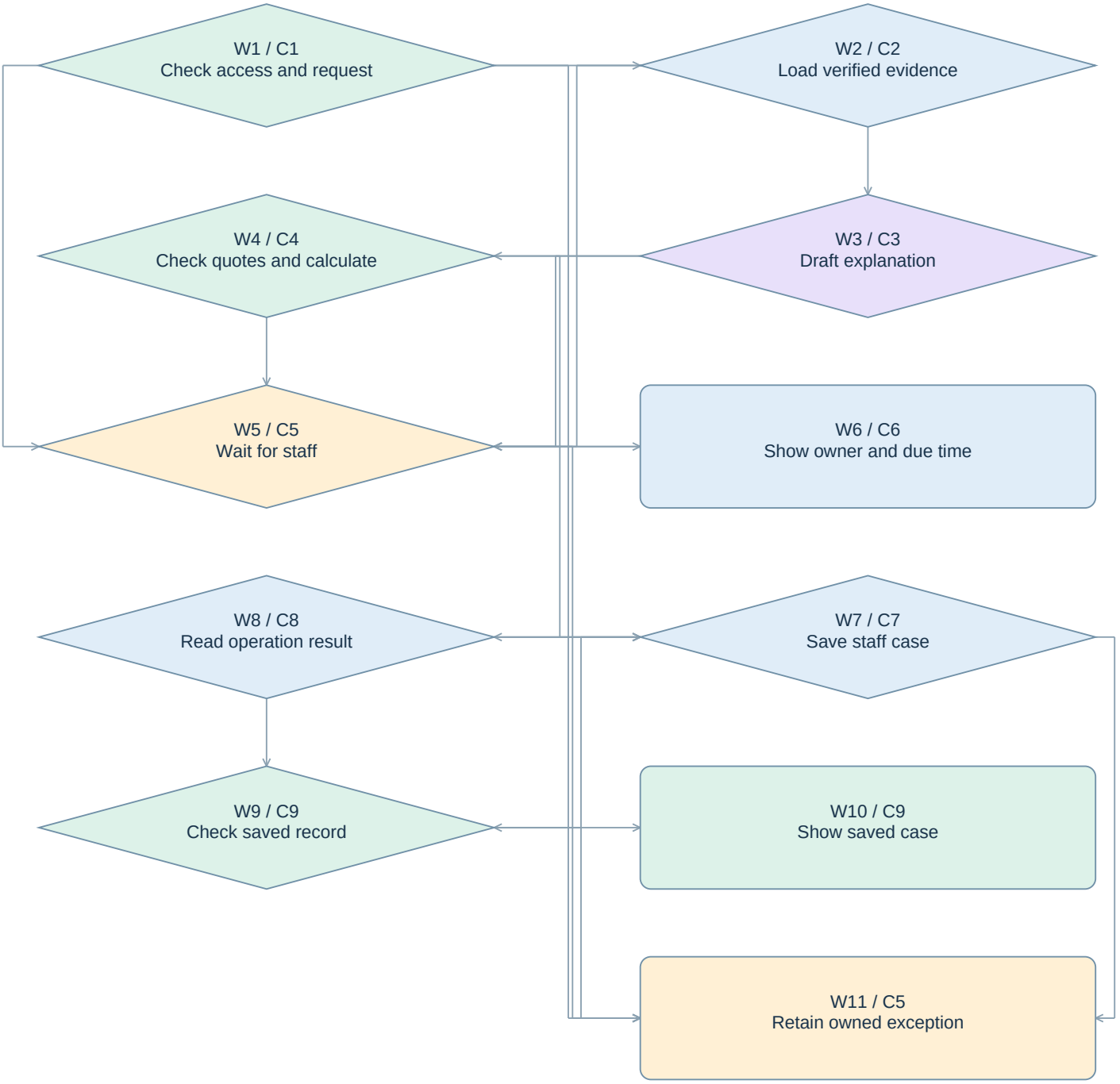
Labels show each leaf mode and risk. Full task inputs, examples and limits are in design.json and PRD.md.

Plain words and scope

Start with draft reviews. Staff decide exceptions and rejections. All claim payments stay with the insurer.

A fingerprint is a check that the approved details have not changed. A record ID is proof from a tool that the change was saved.

Workflow diagram



Diamonds mark choices. Arrows show all routes, including retry and recovery. Exact branch rules are in architecture.md.

Colour	Role
#E8E0FA	Agents (LLM)
#E0EDF8	Code: tools
#DDF2E9	Code: validation
#FFF0D5	Code: human triggers

Top risks and release checks

Risk heat table - severity x occurrence

Each cell names the risks at that severity and occurrence. The full scores, causes, controls and owners are in fmea-register.csv.

S / O	1	2	3	4	5	6	7	8	9	10
10	-	-	FM1	-	-	-	-	-	-	-
9	-	-	FM7	-	FM2, FM3	-	-	-	-	-
8	-	-	FM6	FM4, FM5	FM8	-	-	-	-	-
7	-	-	-	-	-	-	-	-	-	-
6	-	-	-	-	-	-	-	-	-	-
5	-	-	-	-	-	-	-	-	-	-
4	-	-	-	-	-	-	-	-	-	-
3	-	-	-	-	-	-	-	-	-	-
2	-	-	-	-	-	-	-	-	-	-
1	-	-	-	-	-	-	-	-	-	-

Risk	Control summary / full register linked
FM1 Wrong subject medical disclosure	Deny before medical fetch; row-level tenant/member checks and source membership
FM2 Wrong coverage or room adjustment	Signed effective-date precedence, explicit all required rules, integer arithmetic; unknown rules withhold amount
FM3 Forged citation or malicious document	Source-visible OCR admission; exact anchored quote validation; contradictions route review
FM4 Duplicate/concurrent drafts confuse adjudication	Submission identity uniqueness, CAS, semantic candidates require linking
FM5 Uncertain save lost or repeated	Independent readback service, frozen intent, bounded reads; no blind retries
FM6 False completion without proof	Readback verifies case/operation/version and stored bytes before saved display
FM7 Revoked authority still writes	Fresh writer-side RBAC and snapshot version checks; no payment credentials
FM8 Human queue silently stalls	Provisional intake; one-hour pull status; four-hour escalation; 24-hour owned overdue state

Risk scores are proposed estimates. They are not measured odds or risk acceptance.

Components and contracts

This inventory is a summary. Exact typed inputs, outputs and permission rules are in architecture.md and the build milestone files.

Component / group	Responsibility	Task IDs
C1 / Code: validation	Authorize intake	T1
C2 / Code: tools	Verify source evidence	T2
C3 / Agents (LLM)	Explain evidence	T3
C4 / Code: validation	Validate policy and calculate	T3, T4
C5 / Code: human triggers	Own human review	T5, T7
C6 / Code: tools	Show waiting status	T5
C7 / Code: tools	Save staff-facing case	T6
C8 / Code: tools	Reconcile save	T6, T7
C9 / Code: validation	Guard record proof	T7

Shared control boundary

Code checks identity, money, dates and limits. A model may suggest a choice. It cannot authorise or perform a business write.

Architecture and model budget

F1 - Durable code state machine

Select on durable waits, audit, replay and retry control. Database workflow is enough for eleven nodes; a workflow engine is an unverified candidate if timer/replay load grows.

Why: Bounded responsibility with an owned safe outcome.

F2 - Separate records and retrieval

Documents establish evidence; insurer registry establishes policy. Tenant-keyed reads, row isolation, encryption, secrets vault, PHI-free telemetry; deletion/hold and region blocked on H3.

Why: Bounded responsibility with an owned safe outcome.

F3 - Narrow case writer

One case revision blast radius; fresh RBAC/CAS, full-approved details, isolated reconciliation reader. Service owns unresolved operations after session expiry.

Why: Bounded responsibility with an owned safe outcome.

F4 - Provider-neutral private deployment

Compare candidates on region, access boundaries, durable timers, deletion evidence and operating burden with identical tests; no current vendor claims.

Why: Bounded responsibility with an owned safe outcome.

Model cost

Illustrative model cost is USD 0.02112 per case. Unknown prices are not free service. Confirm rates, traffic, staff and tools before launch.

Full call counts, token budgets, rejected choices and costs: design.json, architecture.md and assumptions.md.

Evaluation coverage matrix

Risk / task	EV1	EV2	EV3	EV4	EV5	EV6	EV7	EV8	EV9	EV10	EV11	EV12
FM1	-	X	-	-	-	-	-	-	-	-	-	-
FM2	X	-	X	X	X	-	-	-	-	-	-	-
FM3	-	-	-	-	-	X	X	X	X	-	-	-
FM4	-	-	-	-	-	-	-	-	-	X	X	-
FM5	-	-	-	-	-	-	-	-	-	-	-	X
FM6	-	-	-	-	-	-	-	-	-	-	-	X
FM7	-	-	-	-	-	-	-	-	-	-	-	-
FM8	-	-	-	-	-	-	-	-	-	-	-	-
T1	X	X	-	-	-	-	-	-	-	-	-	-
T2	X	X	-	-	-	X	X	X	X	-	-	-
T3	X	-	-	-	-	X	X	X	X	-	-	-
T4	X	-	X	X	X	X	X	X	X	-	-	-
T5	-	-	-	-	-	-	-	-	-	-	-	-
T6	X	-	-	X	-	-	-	-	X	X	X	X
T7	X	-	-	X	-	-	-	-	-	-	-	X

Evaluation coverage matrix - continued

Risk / task	EV13	EV14	EV15	EV16	EV17	EV18	EV22	EV23	EV24	EV25	EV26	EV27
FM1	-	-	-	-	-	-	-	-	-	-	-	-
FM2	-	-	-	-	-	-	X	X	X	-	-	-
FM3	-	-	-	-	X	-	-	-	-	-	X	-
FM4	-	-	-	-	-	X	-	-	-	-	-	-
FM5	X	-	-	-	-	-	-	-	-	-	-	-
FM6	-	X	-	-	-	-	-	-	-	-	-	-
FM7	-	-	X	-	-	-	-	-	-	X	-	-
FM8	-	-	-	X	-	-	-	-	-	-	-	X
T1	-	-	-	-	-	-	-	-	-	-	-	-
T2	-	-	-	X	X	-	-	-	-	-	X	-
T3	-	-	-	-	X	-	-	-	-	-	X	-
T4	-	-	-	-	-	-	X	X	X	-	-	-
T5	X	-	-	X	-	-	-	-	-	-	-	X
T6	X	-	X	X	-	X	-	X	-	X	-	X
T7	X	X	-	X	-	-	-	-	-	-	-	X

Evaluation coverage matrix - continued

Risk / task	EV28	EV29	EV30	EV31
FM1	-	-	-	-
FM2	-	-	X	X
FM3	-	-	-	-
FM4	-	-	-	-
FM5	-	-	-	-
FM6	-	-	-	-
FM7	X	-	-	-
FM8	-	X	-	-
T1	X	X	-	-
T2	-	-	-	-
T3	-	-	-	-
T4	-	-	X	X
T5	-	X	-	-
T6	-	X	X	-
T7	-	X	-	-

X means the test is linked to that risk or task. It does not mean the test passed. Complete fixtures and assertions: eval-set.json.

Evaluation plan

Test	Layer	Purpose
EV1	golden	Valid room-only recommendation
EV2	golden	Wrong tenant cannot read
EV3	golden	Stale operative policy withheld
EV4	golden	Explicit proportional room rule
EV5	golden	Missing room rule is not invented
EV6	golden	Isolated citation check page
EV7	golden	Isolated citation check clause
EV8	golden	Isolated citation check text
EV9	golden	Hidden instructions excluded
EV10	golden	Two concurrent submissions cannot overwrite
EV11	golden	Exact duplicate returns existing record
EV12	golden	Unknown save survives caller expiry and repeated callback
EV13	golden	Cancellation cannot conceal uncertain save
EV14	golden	Missing record proof cannot show saved
EV15	golden	Revoked authority blocks save
EV16	golden	Missing evidence has a visible owned deadline
EV17	golden	Conflicting source evidence abstains
EV18	golden	Semantic duplicate needs staff linking
EV19	offline	Held-out source and arithmetic validation
EV20	online	Queue and save integrity monitoring
EV21	release	Release regression gate
EV22	golden	Policy boundary uses service date, not current date
EV23	golden	Fresh remaining cover caps the recommendation
EV24	golden	Nonzero copay stays with staff
EV25	golden	Changed payload cannot reuse authorization
EV26	golden	Unverified OCR has no authoritative total
EV27	golden	Reviewed rejection remains staff-facing
EV28	golden	Correction requires original record
EV29	golden	Pre-save cancellation has a proven case record
EV30	golden	Half-up rounding is visible at a minor unit
EV31	golden	Negative charge cannot become a draft

These are test plans. No business controls, provider quality or user outcomes were measured. See eval-set.json for fixtures and pass rules.

Measurement and rollout

R1 - Measure examiner time and decision quality

Measure 2 weeks of eligible case active handling minutes (median per resolved case), draft correctness exact case proportion, and no-PHI breach counts; report strata and unresolved cases separately. H6 target 20% lower median time is hypothesis. Source accuracy...

Why: Bounded responsibility with an owned safe outcome.

R2 - Shadow then assist

Shadow entry H1-H4 verified and EV21 synthetic/adapter gate. Exit after two representative weeks: every safety assertion passes, experts inspect all policy/date/room-rule strata and disagreement reasons. Assist entry shadow exit + staffed queue. Assist exit after four weeks: zero...

Why: Bounded responsibility with an owned safe outcome.

R3 - Test handling-time benefit

Cluster-randomize examiner teams to assistant/control; stratify by plan, room cap and document type. Keep teams in one arm to reduce contamination. Both arms retain same human approval. Four-week observation minimum after baseline; analytics calculates continuous-time variance, chosen minimum...

Why: Bounded responsibility with an owned safe outcome.

R4 - Own rollout and unresolved cases

Director signs expansion only after EV21 and H1-H4 evidence; privacy/rules/security owners can veto safety. Operations owns unresolved save tickets across caller expiry, reviews every four hours, escalates to incident lead at deadline. Human approves any future settlement scope...

Why: Bounded responsibility with an owned safe outcome.

Build handoff and review

Start with build/00-overview.md. Load one milestone file at a time. Exact source rules remain in design.json; do not treat design approval as live write permission.

X1: Implement declared staff-case workflow only. No operational API support or medical/insurance advice is established.

Review dimensions - evidence in review.json

Dimension	Score	Dimension	Score
D1	3/4	D7	3/4
D2	2/4	D8	4/4
D3	3/4	D9	3/4
D4	3/4	D10	2/4
D5	3/4	D11	3/4
D6	4/4	D12	2/4

IR1 major OPEN: C4 requires a fresh authoritative accumulator version and explicitly sends a missing version to review. Positive initial-analysis fixtures EV1, EV4, EV9, EV30 omit accumulatorVersion...

IR2 major OPEN: The concurrency fixture starts operations OP-A and OP-B, but its only immutable authorized operation payload/hash is for OP-1. Under C7 exact operation/resource binding, neither...

Glossary

canonical: agreed

discriminated: typed

payload binding: approved details

digest: fingerprint

idempotency: duplicate-write protection

reciprocal: two-way

terminal disposition: final outcome

calc_digest: a fingerprint of the approved amount

After delivering DESIGN.pdf, ask the user: Would you also like a visual HTML version in the same ten-stage structure as the app? Generate HTML only after opt-in, unless already requested.

Full assumptions, owners and evidence: assumptions.md. Review method and failed rounds are preserved in review.json and review-round-*.json.